

IEEE 802.3 MAC Protocol

⊕ 1-Persistent CSMA/CD with Truncated Binary Exponential Backoff

➡ Collision resolution: Truncated Binary Exponential Backoff

- If a station has experienced the n^{th} failure in a row for a frame, it selects an integer value (K) at random from $\{0, 1, \dots, 2^m-1\}$ where $m = \min(n, 10)$ and waits for K mini-slots (each mini-slot = $2t_{\text{prop}}$) before sensing again
- The increasing range of selection for backoff after each collision is intended to increase the likelihood that re-transmission will succeed
- Up to 16 re-transmission attempts will be allowed, after which the system gives up

CPR E 4890 -- D.Q.

Minimum Frame Size in IEEE 802.3 Ethernet

- ⊕ **Observation:** the transmitter must keep transmitting (i.e., hold the carrier) for the entire $2t_{\text{prop}}$ period, in order to detect whether its own frame is involved in a collision

⊕ Example:

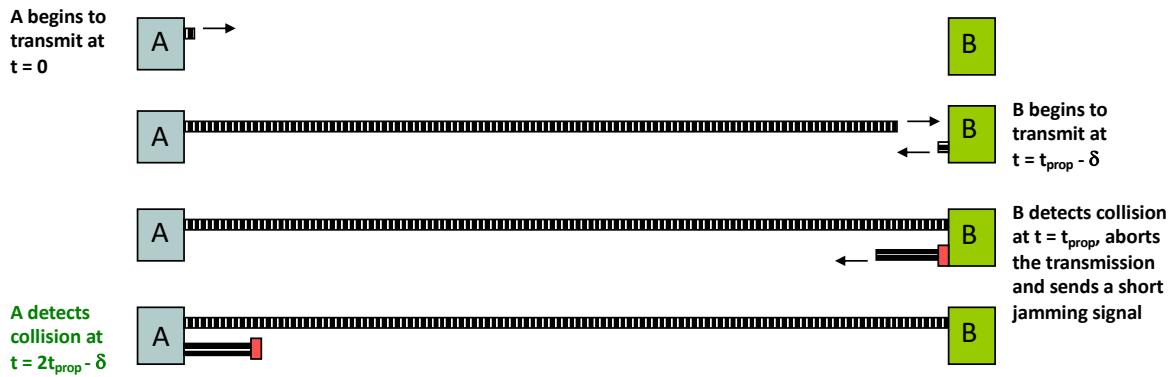
- ➡ Transmission rate = 10 Mbps
- ➡ Maximum distance = 2500 m (500 m segments * 5) + 4 repeaters
- ➡ Mini-slot time = $2t_{\text{prop}} = 2 \times [2500 \text{ m} / (2 \times 10^8 \text{ m/s}) + 3 \mu\text{s} \times 4] = 49 \mu\text{s}$
 - Absolute minimum frame size is $49 \mu\text{s} \times 10 \text{ Mbps} = 490 \text{ bits}$
- ➡ IEEE 802.3 standard requires 64 bytes = 512 bits

- ⊕ So, when the transmission rate increases, for CSMA/CD to operate correctly, we need to

- ➡ Increase the minimum frame size, or
- ➡ Reduce the maximum distance between two stations

CPR E 4890 -- D.Q.

CSMA/CD Reaction Time



- It takes A **up to $2t_{prop}$** time to find out whether it has captured the channel successfully